MSG No. 192.245 $P6/m'cc$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `62'2'`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,2,3,4,5,6,13,14,15,16,17,18]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[7,8,9,10,11,12,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `6/m'..`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,3,5,16,17,18]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[2,4,6,13,14,15]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[7,8,9,20,22,24]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[10,11,12,19,21,23]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,3,5,20,22,24]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[2,4,6,19,21,23]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[7,8,9,16,17,18]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `6..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3,4,5,6]`
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[7,8,9,10,11,12]`
3	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[13,14,15,16,17,18]`
4	$[0,\, 0,\, -z]$	`[19,20,21,22,23,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, \frac{1}{4}\right]$	`[1,4,13,17]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$	`[2,5,15,16]`
3	$\left[0,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[3,6,14,18]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,11,19,22]
5	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[8,12,21,24]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[9,10,20,23]

Table 7: Wyckoff site: 6g, site symmetry: 2/m'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,19,22]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,20,23]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,6,21,24]
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[7,11,13,17]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[8,12,14,18]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,15,16]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2,4,6]
3	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[7,8,9]
4	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[10,11,12]
5	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[13,14,15]
6	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[16,17,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[19,21,23]
8	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[20,22,24]

Table 9: Wyckoff site: 12i, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4]
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[2,5]
3	$\left[0, \frac{1}{2}, z\right]$	[3,6]
4	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[7,11]
5	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[8,12]
6	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[9,10]
7	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[13,17]
8	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[14,18]
9	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[15,16]
10	$\left[\frac{1}{2}, 0, -z\right]$	[19,22]
11	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[20,23]
12	$\left[0, \frac{1}{2}, -z\right]$	[21,24]

Table 10: Wyckoff site: `12j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,13]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,16]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,14]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,17]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,18]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,19]`
8	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[8,21]`
9	$\left[x,\ x,\ \frac{3}{4}\right]$	`[9,23]`
10	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[10,20]`
11	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[11,22]`
12	$\left[0,\ x,\ \frac{3}{4}\right]$	`[12,24]`

Table 11: Wyckoff site: `12k`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,17]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,15]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,18]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,13]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,16]`
6	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[6,14]`
7	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[7,22]`
8	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[8,24]`
9	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[9,20]`
10	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[10,23]`
11	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[11,19]`
12	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[12,21]`

Table 12: Wyckoff site: `12l`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,22]`
2	$[x-y,\ x,\ 0]$	`[2,23]`
3	$[-y,\ x-y,\ 0]$	`[3,24]`
4	$[-x,\ -y,\ 0]$	`[4,19]`
5	$[-x+y,\ -x,\ 0]$	`[5,20]`
6	$[y,\ -x+y,\ 0]$	`[6,21]`
7	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[7,17]`
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[8,18]`
9	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[9,16]`
10	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[10,15]`
11	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[11,13]`
12	$\left[y,\ x,\ \frac{1}{2}\right]$	`[12,14]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[x-y,\, x,\, z]$	`[2]`
3	$[-y,\, x-y,\, z]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[-x+y,\, -x,\, z]$	`[5]`
6	$[y,\, -x+y,\, z]$	`[6]`
7	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[7]`
8	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[8]`
9	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	`[11]`
12	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[12]`
13	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[y,\, x,\, \frac{1}{2}-z\right]$	`[14]`
15	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	`[15]`
16	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	`[16]`
17	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[17]`
18	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	`[18]`
19	$[-x,\, -y,\, -z]$	`[19]`
20	$[-x+y,\, -x,\, -z]$	`[20]`
21	$[y,\, -x+y,\, -z]$	`[21]`
22	$[x,\, y,\, -z]$	`[22]`
23	$[x-y,\, x,\, -z]$	`[23]`
24	$[-y,\, x-y,\, -z]$	`[24]`